MILITARY SPECIFICATION

FORCEPS, HEMOSTATIC, HALF-CURVED, MIXTER, 7-1/2 INCH

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type and size of hemostatic forceps of the Mixter design, suitable for use in surgical procedures.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

GG-I-526	Instruments, Dental and Surgical; General Specifications.
MMM-A-250	Adhesive, Water-Resistant, (For Sealing Fiberboard Boxes).
PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.
PPP-T-76	Tape, Pressure Sensitive Adhesive Paper (For Carton Sealing).

Military

MIL-P-116	Preservation, Methods of.
MIL-L-10547	Liners, Case, and Sheet Overwrap; Water-Vaporproof or Waterproof, Flexible.

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

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(Copies of specifications and standards, required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General Requirements. The forceps shall be in accordance with the applicable requirements of GG-I-526 and the following option exercises under that specification:

3.2 Material. The forceps shall be fabricated from Class 4, Type 410 or 420 material.

3.3 Hardness. After proper stress relieving, the forceps shall have a Rockwell hardness of C40 to C45, with the exception of the pin.

3.4 Style, design and dimensions. The forceps shall be of the Mixter design, with curved tip, suitable for use in surgical procedures. Shall be in accordance with Figure 1.

3.4.1 Locks. Shall be box type lock.

3.5 Finish. Finish shall be mirror. Locks and ratchets shall be smooth and shall be polished, but need not be buffed to a mirror finish.

3.6 Lock stiffness. Forceps of box lock construction shall have shanks tested for lock stiffness in accordance with GG-I-526.

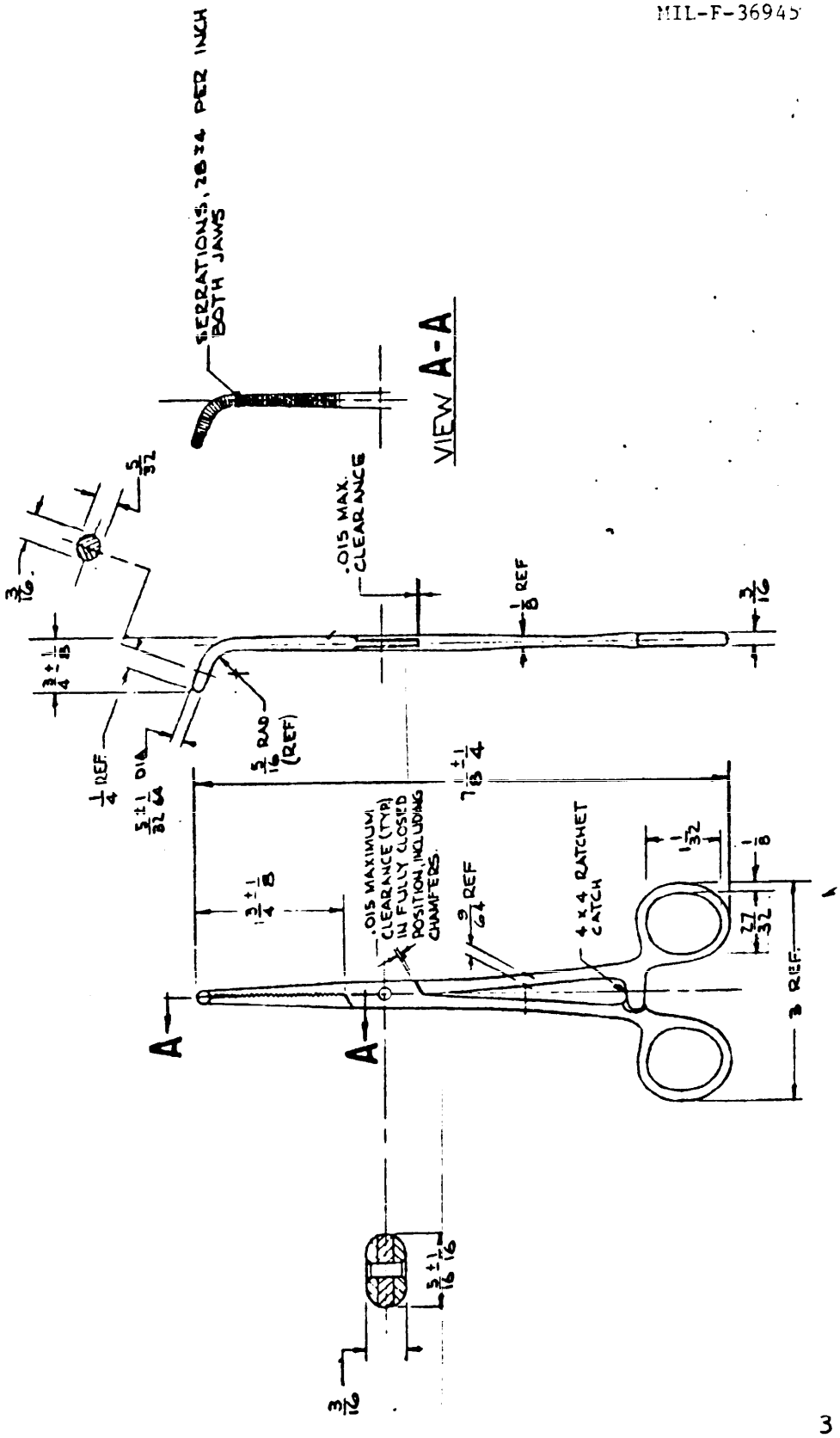
3.7 Identification marking. In addition to GG-I-526 requirements, forceps of foreign manufacture shall also be marked "stainless".

3.8 Jaw and ratchet operation. When tested as specified in 4.4.1, the jaw tip shall close first and progressively tighten with each ratchet position. Each ratchet position shall hold firmly and release instantly.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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NOTES:
TOLERANCES INDICATED DELINEATE ACCEPTABLE EXTREME DIMENSIONS, AND ARE NOT INTENDED TO PERMIT VARIATIONS WITHIN A SINGLE INSTRUMENT WHICH SACRIFICE UNIFORMITY, SYMMETRY AND FIRST QUALITY WORKMANSHIP.
UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES AND TOLERANCE IS ± 1/32 IN.

FIGURE 1

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4.1.1 Inspection. Inspection shall be in accordance with GG-I-526.

4.1.2 Records. Records of examination and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.3 Certificates of quality. Certificates of quality supplied by the manufacturer of the corrosion resisting steel, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificates shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the Item Identification, the name of the material, the lot number, the lot size, the sample size, the date of testing, the test method, the individual test results, and the specification requirements.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and as indicated in Table I. The unit of product for examination purposes shall be one hemostatic forceps.

Table I. Sampling Criteria for Examination

Type of defect	Inspection level	AQL (Percent defective)
Major	II	1.0
Minor	II	2.5

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105, and as indicated in Table II. Unit of product for test purposes shall be one forceps.

Table II. Sampling Criteria for Tests

Component	Characteristic	Requirement	Test procedure	Inspection level	AQL
1-Forceps	Composition analysis	3.2 and GG-I-526	GG-I-526	**	*
	Hardness	3.3 and GG-I-526	GG-I-526	S-2	*
	Dimensions	3.4	--------	S-2	4.0
	Boil test	GG-I-526	GG-I-526	S-2	*

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Table II. Sampling Criteria for Tests (Cont'd.)

Component	Characteristic	Requirement	Test procedure	Inspection level	AQL
1-Forceps	Copper sulfate	GG-I-526	GG-I-526	S-2	*
	Box lock stiffness	3.6	GG-I-526	II	6.5
	Distortion test	GG-I-526	GG-I-526	S-2	*
	Jaw and Ratchet Operation test	3.8	4.4.1	S-2	1.0

* Acceptance number shall be zero for all sample sizes.
** Four ounces of each lot of metal used in the construction of the forceps shall be selected for chemical analysis. Where feasible, the same sample unit shall be used for the determination of two or more test characteristics.

4.3 Examination. The forceps shall be examined to determine compliance with all the requirements contained in this specification. Examination shall be conducted as specified in Table I and III. Examination however, shall not be restricted to defects listed in Table III. Any forceps in the sample containing one or more defects shall be rejected, and if the number of defective items in any sample exceeds the acceptance number for test sample, the lot represented by the sample shall be rejected.

Table III. Classification of Defects

Categories	Defects
Critical	None defined.
Major	
101	Style and design not as specified.
102	Forceps not free from cracks, pits, dents, nicks, sharp edges and burrs.
103	Serrations not as specified.
104	Ratchet teeth not uniform, or function poorly.
105	Forceps not of box lock design.
106	Forceps not free from excessive play in box lock area.
Minor	
201	Finish not mirror, and not uniform.
202	Identification marking not present, correct, complete or legible.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

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4.4.1 Jaw and operation test. The forceps shall be positioned to the first ratchet position, but not locked. Forceps with jaw tips not touching shall be rejected. Forceps that do not progressively tighten with each ratchet advance shall be rejected.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Unit of issue. One forceps as specified, constitutes one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in Table IV shall be packaged in each unit, intermediate, and exterior container as applicable, for the required level of protection specified in the procurement document.

Table IV. Packaging Quantities

Unit package	Intermediate package	Exterior container
1 unit	12 units	288 units

5.1.2.1 Packing variation permitted. If the required number of units in the entire shipment is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

5.1.3 Level A.

5.1.3.1 Cleaning and drying. Each unit shall be thoroughly cleaned and dried in accordance with the applicable method as outlined in MIL-P-116.

5.1.3.2 Preservative. None.

5.1.3.3 Intimate wrap. Each unit shall be wrapped in neutral tissue or other suitable anticorrosive material.

5.1.3.4 Unit package. Each unit shall be packaged in accordance with MIL-P-116, method IC-1.

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5.1.3.5 Intermediate package. Intermediate package shall be a box of appropriate size and design constructed in accordance with PPP-B-636, type CF, class weather-resistant. Boxes shall be closed by taping the top and bottom flaps with 2 inch wide tape conforming to PPP-T-76. Closing of bottom flaps with adhesive conforming to MMM-A-250 or metal stitches is permitted.

5.1.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be Level A, B or C as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant, grade V3c. Closure and strapping shall be as specified in the appendix of the applicable box specification. Special requirements of applicable table shall apply for PPP-B-636 box.

5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification. Closure of fiberboard boxes shall conform to method II as specified in PPP-B-636. In addition, fiberboard boxes shall conform to the special requirements as specified in PPP-B-636.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Interior packages. Each unit package and each intermediate package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number and name of contractor shall be shown.

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5.3.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The hemostatic forceps are to be used in general surgical procedures.

6.2 Ordering data. Procurement documents should specify the following:

(a) Number, title and date of this specification.
(b) Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

6.3 National Stock Number (NSN) coverage. This specification covers the following item listed in the Federal Supply Catalog:

National Stock Number	Item Identification
6515-00-065-3181	FORCEPS, HEMOSTATIC, Half Curved, Mixter 7-1/2 inch

It does not include all types or sizes as indicated by the title nor all types which are commercially available.

6.4 Supersession data. This specification includes the requirements of Purchase Description No. 3, dated 6 February 1968 for NSN 6515-00-065-3181 FORCEPS, HEMOSTATIC, Half Curved, Mixter, 7-1/2 inch.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6515-1698

Review activities:
Army - MD
Navy - MS
Air Force - 03

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

MIL-F-36945

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

   A. GIVE PARAGRAPH NUMBER AND WORDING

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*	TELEPHONE NO.
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